

Introduction

Wood is one of the oldest natural materials utilized by humanity. Due to its excellent structural properties, affordability, and widespread availability in the market, it is extensively used in both the construction and furniture industries. Therefore, this project presents the design and architectural scale model of a **Two-Storeyed Timber Building**, highlighting the strength, sustainability, and aesthetic potential of timber construction.

Objectives

The objective of this project is to gain a clear understanding of the general characteristics and properties of timber and to design a structurally sound **Two-Storeyed Timber Building** by studying the interconnections and integration of its essential structural components.

General Characteristics of Timber

Thousands of distinct tree species exist worldwide, each possessing unique qualities and structural properties. In general, timber is broadly categorized into two major types: **Softwood** and **Hardwood**.

Softwood

Softwoods are generally characterized by their soft wood texture and needle-like or scale-like leaves. These timbers are derived from fast-growing trees. Due to their softness, softwoods are easy to work with and can be conveniently bent and shaped for various applications. Example: **Pine Tree**



Pine Tree (Softwood)

Hardwood

Hardwoods possess dense and hard wood textures, harvested from slow-growing trees. Due to their superior strength, high durability, and overall quality, they are widely utilized in major construction projects. Hardwoods offer higher hardness and are generally superior in performance compared to softwoods. Example: **Teak**



Teak (Hardwood)

Commonly Used Timber Types

The commonly used timber types in Myanmar's construction industry are as follows:

- **Teak:** A premium hardwood species known for its remarkable longevity and excellent water-resistant properties.

- **Pyinkado (Ironwood):** An extremely dense and high-strength timber with superior resistance to termite attacks.
- **Padauk:** One of the strongest timber types in Southeast Asia. Naturally durable, it boasts a lifespan of at least 12 years even without preservative treatments.
- **Kanyin:** Derived from massive trees that reach heights of 70 to 100 feet, providing large-volume structural timber.

Building Structural Components

To construct a stable and resilient structure, key components must be effectively connected. The primary structural components of a two-storey timber building are:

- **Roof**
- **Column**
- **Beam**
- **Joist**



Structural Components Layout

Design of a Two-Storeyed Timber Building

When designing a two-storey timber building, the process involves the following structural steps:

- (1) Roof Design - Calculate the forces acting on the roof and select the appropriate truss sizes capable of safely withstanding these loads.
- (2) Floor Joist - Design Calculate the Dead Load acting on the floor and select the appropriate Live Load based on the **IBC-2006** code standards to design and size the floor joists accordingly.
- (3) Beam Design - Determine and select beam sizes that can safely resist and support the loads transferred from the floor joists.
- (4) Column Design - Calculate and select column dimensions capable of supporting the cumulative structural loads transferred from the beams.

Member Sizes Obtained from Structural Design

No	Members	Sizes
1	Purlin	3inch x 3inch
2	Rafter	3inch x 6inch
3	Joist	3inch x 6inch
4	Beam	6inch x 8inch
5	Column	5inch x 5inch

Fabrication of Two-Storeyed House Scale Model



Left Elevation



Back Elevation



3D View



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Design of Two Storeyed Timber Building

